



TAILSCALE OVERLAY – P2P MESH

all nodes on the same network · NAT traversal · encrypted point-to-point

NODE_ALPHA · Razr Ultra
SM8750 · 16 GB · ~45 TOPS
Hexagon HTP v79

- Horizons UI (lifecycle core)
- novus-agenti · NovA-Claw
- CCR on-device sessions

Tailscale · WiFi

NODE_BETA · Jetson Orin Nano Super
8 GB · 512 GB DDR4 SSD · ~67 TOPS
MAIN COMPUTE HUB · headless

- peer-agent (Hydra rotation)
- utilities-agent (housekeeping)
- inference server

HIGH-SPEED DATA CABLE (wired · direct)

NODE_GAMMA · Rubik Pi 3 / DragonWing
8 GB · ~128 GB SSD · ~14 TOPS
VISUAL OS · dual monitor · keyboard

- desk display (VNC server)
- DragonWing SoC on-board

Tailscale · WiFi (to ALPHA / mesh)

THIN CLIENT (no compute node)
Galaxy Tab S9 FE+ · Wi-Fi only

- SSH → NODE_BETA (headless)
- AVNC → NODE_BETA (desktop VNC)
- Termux · SD card (Jetson flash)

NODE_ALPHA ACTIVE Motorola Razr Ultra 2025 <table><tr><td>CHIP</td><td>SM8750 · Snapdragon 8 Elite</td></tr><tr><td>RAM</td><td>16 GB</td></tr><tr><td>STORAGE</td><td>~512 GB</td></tr><tr><td>AI TOPS</td><td>~45 TOPS</td></tr><tr><td>NPU</td><td>Hexagon HTP v79</td></tr><tr><td>FORM</td><td>Phone · primary user node</td></tr></table> SERVICES RUNNING · Horizons UI – lifecycle core; nothing lives without it · novus-agenti – phone agent, runtime manifests · NovA-Claw – execution surface (nova-skills) · CCR – claude.ai UI · on-device execution · ONNX/ORT + Sherpa-ONNX voice layer (in-process) · QAIRT/HTP inference (when models offloaded + bare metal)	CHIP	SM8750 · Snapdragon 8 Elite	RAM	16 GB	STORAGE	~512 GB	AI TOPS	~45 TOPS	NPU	Hexagon HTP v79	FORM	Phone · primary user node	NODE_BETA CONFIGURING Jetson Orin Nano Super <table><tr><td>CHIP</td><td>Jetson Orin Nano Super</td></tr><tr><td>RAM</td><td>8 GB</td></tr><tr><td>STORAGE</td><td>512 GB DDR4 SSD (added)</td></tr><tr><td>AI TOPS</td><td>~67 TOPS</td></tr><tr><td>FORM</td><td>Headless · main compute hub</td></tr><tr><td>CONNECT</td><td>Tailscale WiFi ← ALPHA; wired → GAMMA</td></tr></table> SERVICES RUNNING · peer-agent – twins with novus-agenti; Hydra cloud rotation · utilities-agent – housekeeping, load/unload operator agents · inference server – swappable backend (never hardcoded) · AVNC server – portable desktop from Galaxy Tab · GCP pipeline endpoint (files-inference-node secretary)	CHIP	Jetson Orin Nano Super	RAM	8 GB	STORAGE	512 GB DDR4 SSD (added)	AI TOPS	~67 TOPS	FORM	Headless · main compute hub	CONNECT	Tailscale WiFi ← ALPHA; wired → GAMMA
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LOOPBACK SHARED

`127.0.0.1:8080` bound by app daemon · Termux shares loopback · NPU access no inversion needed

INBOUND LISTENER

App needs listener for mic/voice/WebView-OAuth from Termux – not yet built **ABSENT**

CONFIG LOCATION Network config lives in `aesop-xi-protocol/` – infrastructure/pathways layer

VOICE / NPU PATHWAY

OPERATOR CANON · AUG 2 2026

OPERATOR – NEURO MESH SESSION · AUG 2 2026

"my APK already ships on device with an stt / TTS layer that's running a ONNX/ORT through a llama server. this also allows the model inside of my Termux to utilize the QAIRT model path that allows a GGUF model to run through llama to ggml to kotlin kernel and librc runtimes to land directly on the NPU through the HTP SDK net pathway – I said that pathway will only be utilized when my on device models or my on-device agent through the Horizons UI is offloaded and the APK is running bare metal"

STT / TTS (in-process · always)

ONNX/ORT — llama server — Sherpa-ONNX

Moonshine STT (in-process on AAR) · Kokoro TTS (in-process)

NPU CHAIN (only when Horizons UI offloaded + bare metal)

GGUF → llama → ggml → Kotlin kernel → librc runtimes → Hexagon HTP SDK → NPU

QAIRT model path · HTP pathway · Snapdragon 8 Elite · ~45 TOPS

DESIGN CONSTRAINT

never hardcode inference backend – keep swappable across all phases

[STT] **BROKEN MID-CHAIN** – points at `127.0.0.1:8091` , nothing binds it. Fix: Moonshine in-process on AAR already shipping **UNVERIFIED ON DEVICE**

[TTS] **works in-process** – Sherpa AAR · Kokoro

[LLM] **falls to llmRuntime.streamAudio** – looks like model problem, is STT problem

CCR ON-DEVICE EXECUTION

TERMUX → CLAUDE.AI

NODE_ALPHA (Razr Ultra)

Termux — start `claude` ————— session spawns

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claude.ai = user interface (phone browser / app)

execution environment = on-device container (not cloud)

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repo access · Drive MCP · GitHub MCP · all tool calls run on device

When `claude` runs in Termux, the session is an on-device container. The user's phone is both the UI and the execution host. Drive MCP, file reads/writes, git pushes – all run on NODE_ALPHA. The `claude.ai` web interface is just the chat layer.

THIN CLIENT – NO COMPUTE NODE		SETUP DOCS IN <code>home-node/jetson-orin/</code> · <code>termux-environment/</code>
DEVICE	Samsung Galaxy Tab S9 FE+ · Wi-Fi only · no SIM	
ROLE	SSH into NODE_BETA · AVNC into NODE_BETA for portable desktop from couch	
CONNECT	NODE_BETA only – Rubik Pi 3 handles desk, Tab handles couch/portable via VNC	
TOOLS	Termux (installed) · SD card used to flash Jetson Nano	
CONFIG DOCS	<code>home-node/jetson-orin/</code> (VNC server config) · <code>termux-environment/</code> (AVNC client setup)	

MESH TOTALS	REPOS ON THIS STACK
TOTAL AI TOPS	horizons-ui
TOTAL RAM	<code>c10vis-poem/Horizons-UI</code>
TOTAL STORAGE	novus-agenti
~1.1 TB (SSD + NVMe + flash)	<code>c10vis-poem/novus-agenti</code>
COMPUTE NODES	aesop protocol <code>c10vis-poem/aesop</code>
3 active · all P2P servers	nova-skills
THIN CLIENTS	<code>c10vis-poem/nova-skills</code>
1 · Galaxy Tab S9 FE+	home-node/
	Jetson + DragonWing · configs
	<code>termux-env/</code> Termux scripts · CLI
	node-gamma-rubik-pi
	device node (name confirmed)

BUILD ROADMAP (IN ORDER)		DESIGN CONSTRAINT: NEVER HARDCODE INFERENCE BACKEND
1	Home node running on all cylinders CURRENT PHASE NODE_BETA + NODE_GAMMA configured · Tailscale mesh stable · voice layer closed (Moonshine in-process) · Termux backend listener built · runtime params first-class	
2	AEC application – communications + safety layer DESIGNED Communications workflow · software bridging · safety layer protocol. Target: prototype in 6-8 months	
3	ARM64 + commercial GPU DESIGNED Modular TBD form factor – PCIe attach or external enclosure. GOAL: home node replaces the laptop. Operator's own custom design.	
4	Broader modular hardware ecosystem PRIVATE Metamaterials designs · novel thermal management systems. At least 2 potentially patentable designs. ⚠ NOT documented in any public or shared repo until filed.	

[ALPHA] **built-verified** Razr Ultra hardware + Horizons UI APK
[ALPHA] **built-verified** Sherpa-ONNX Kokoro TTS in-process
[ALPHA] **built-verified** Tailscale node – mesh enrolled
[ALPHA] **built-unverified** Moonshine STT in-process on AAR (CI green · not device-tested)
[ALPHA] **built-unverified** NPU HTP pathway (QAIRT/HTP chain · bare metal only)
[ALPHA] **absent** Termux inbound listener (mic/voice/OAuth)
[ALPHA] **absent** runtime params first-class (temperature/verbosity/cores)
[BETA] **built-verified** Jetson Orin Nano Super hardware + DDR4 SSD
[BETA] **built-unverified** Tailscale enrollment · mesh connectivity to ALPHA
[BETA] **absent** peer-agent deployment
[BETA] **absent** utilities-agent deployment
[BETA] **absent** AVNC server config (for Tab access)
[GAMMA] **built-verified** Rubik Pi 3 hardware · DragonWing SoC
[GAMMA] **built-unverified** high-speed data cable to BETA
[GAMMA] **absent** visual OS fully configured · dual monitor active
[NET] **built-verified** Tailscale overlay – all three nodes enrolled
[NET] **absent** full mesh routing verified (all paths tested)